

SMART LEAVE APPROVAL SYSTEM WITH AI-POWERED DECISION SUPPORT

Project Documentation

Project Title: **Smart Leave Approval System with AI-Powered Decision**

Objective

The objective of the Smart Leave Approval System project is to automate and streamline the employee leave management process within an organization. The application enables employees to submit leave requests, allows managers to review and approve or reject requests, automatically calculates leave duration, performs AI-based risk assessment, and generates intelligent recommendations to support decision-making.

The system also provides role-based access control, automated approval workflows, real-time dashboards, and analytical reports for monitoring leave activities. By leveraging the ServiceNow platform, the application centralizes leave management, reduces manual effort, improves transparency, enhances operational efficiency, and supports managers with AI-powered insights during the leave approval process.

The Smart Leave Approval System serves as a secure and scalable solution for managing employee leave requests while providing better visibility into leave trends, approval status, and organizational workforce planning.

Milestone 1: Requirement Analysis – Step-by-Step Process

Step 1: Identify the Problem

Analyze the existing leave management process.

Problem Statement:

- Employees submit leave requests manually.
- Managers track requests through emails or spreadsheets.
- Approval process is slow.
- No centralized system for monitoring leave requests.

Step 2: Define Project Objective

Objective:

Develop a Smart Leave Approval System in ServiceNow to automate leave request submission, approval, tracking, and reporting.

Step 3: Identify Stakeholders

List all users involved in the system:

Stakeholder	Responsibility
Employee	Submit leave requests
Manager	Approve or reject requests
Administrator	Manage application and users

Step 4: Gather Functional Requirements

Employee Requirements

- Apply for leave.
- View leave request status.
- Track leave history.

Manager Requirements

- View leave requests.
- Approve leave requests.
- Reject leave requests.

Administrator Requirements

- Manage users and roles.
- Monitor reports and dashboards.

Step 5: Gather Non-Functional Requirements

- Secure access using roles.
- Fast response time.
- User-friendly interface.
- Reliable data storage.
- Scalable application design.

Step 6: Define System Features

Create a list of features to be implemented:

1. Leave Request Form

2. Leave Approval Workflow
3. Total Days Calculation
4. AI Risk Assessment
5. AI Recommendation
6. Role-Based Security
7. Dashboard and Reports

Step 7: Create Use Cases

Use Case 1: Apply Leave

Actor: Employee

Steps:

1. Open Apply Leave module.
2. Fill leave details.
3. Submit request.

Output:

Leave request created.

Use Case 2: Approve Leave

Actor: Manager

Steps:

1. Open pending requests.
2. Review request.
3. Approve or reject.

Output:

Status updated.

Step 8: Define AI Requirements

AI functionality included:

AI Risk Assessment

- Low Risk (1–3 days)
- Medium Risk (4–7 days)
- High Risk (>7 days)

AI Recommendation

- Normal request
- Manager review recommended
- Extended leave requires additional review

Step 9: Define Security Requirements

Create roles:

x_1877519_smart_0.employee
x_1877519_smart_0.leave_manager
admin

Permissions:

Role	Permission
Employee	Create and view own requests
Manager	Approve/reject requests
Admin	Full access

Step 10: Document Expected Outcomes

Expected benefits:

- Reduced manual work
- Faster approval process
- Better transparency
- Improved decision-making using AI risk assessment
- Real-time analytics through dashboards